Literature-implied negative answer for arXiv:1407.1065

The untrimmed Wirtinger Flow initializer cannot use $m = O(n)$ samples

Status. literature_implied_answer (full negative answer to the stated linear-sampling question). This is not a new run counterexample. A later paper identifies the extreme-measurement obstruction; the short argument below makes its implication for the source's 1/8 basin explicit.

Source question. In Candès, Li, and Soltanolkotabi, *Phase Retrieval via Wirtinger Flow: Theory and Algorithms*, arXiv:1407.1065, PDF page 6, Theorem 3.3 proves the Gaussian-model guarantee with $m \geq c_0 n \log n$. Immediately afterward the authors observe that injectivity needs only linearly many samples and ask whether Theorem 3.3 holds when m is merely proportional to n.

The theorem uses the leading eigenvector of

$$Y = \frac{1}{m}\sum_{r=1}^{m} y_r a_r a_r^*, \qquad y_r = |\langle a_r, x\rangle|^2, \tag{1}$$

scaled to have norm asymptotic to $\|x\|$, and requires $\operatorname{dist}(z_0, x) \leq \|x\|/8$.

Supporting literature. Chi, Lu, and Chen, *Nonconvex Optimization Meets Low-Rank Matrix Factorization: An Overview*, arXiv:1809.09573, Section 8.3.1 (PDF page 40), cites the standard phase-retrieval spectral construction and proves

$$\|Y\| \geq \frac{(\max_r y_r)\|a_{r_*}\|^2}{m} \asymp \frac{n \log m}{m}. \tag{2}$$

They conclude that this is incompatible with linear sample complexity and that the standard recipe must be modified by truncating the large measurements. Their Theorem 27 records the resulting $m = O(n)$ guarantee for the *truncated* spectral estimator, not the source initializer.

Identification and rigorous bridge. Normalize $\|x\| = 1$ and take $m/n \to \alpha \in (0, \infty)$. Formula (2) tends to infinity in probability (for complex Gaussian vectors, y_r is exponential; the same maximum argument applies). On the other hand,

$$x^* Y x = \frac{1}{m}\sum_{r=1}^{m} y_r^2 \longrightarrow \mathbb{E} y_1^2 = 2, \tag{3}$$

and either normalization in the source makes $\|z_0\|^2 \to 1$.

Let $\Lambda = \|Y\|$ and let z_0 be a scaled leading eigenvector. If, after a phase rotation, $z_0 = x + h$ with $\|h\| \leq 1/8$, positivity of Y and $|u+v|^2 \leq 2|u|^2 + 2|v|^2$ give

$$\|z_0\|^2 \Lambda = z_0^* Y z_0 \leq 2x^* Y x + 2h^* Y h \leq 2x^* Y x + \frac{1}{32}\Lambda. \tag{4}$$

With probability tending to one, $\|z_0\|^2 \geq 1/2$ and $x^* Y x \leq 3$, so (4) would force $\Lambda \leq 64/5$. This contradicts (2). Thus

$$\Pr\{\operatorname{dist}(z_0, x) \leq 1/8\} \longrightarrow 0. \tag{5}$$

Claim. *For every fixed sampling ratio $\alpha > 0$, the untrimmed spectral initializer in arXiv:1407.1065 fails the initialization conclusion of Theorem 3.3 with probability tending to one when $m/n \to \alpha$. Consequently Theorem 3.3 cannot hold in the proposed linear-sampling regime with that initializer.*

For completeness, (2) itself follows by choosing an index with $y_r \geq \frac{1}{2}\log n$ (such an index exists with probability tending to one) and using $\|a_r\|^2 \geq n/2$. The probability of the first event is at least $1 - \exp(-\Theta(\sqrt{n}))$ and the second fails with exponentially small probability. Hence $\Lambda \geq c_\alpha \log n$.

Scope. This does not rule out linear-sample phase retrieval or modified Wirtinger Flow. Truncated Wirtinger Flow and other preprocessed spectral methods do achieve $m = O(n)$. It rules out exactly the source theorem in the proposed regime with its original untrimmed spectral initialization.

Search evidence. The run indexes contained no prior packet for arXiv:1407.1065. Exact-question, initializer, truncation, and extreme-measurement searches located arXiv:1809.09573. Later phase-transition papers analyze bounded preprocessing and likewise motivate truncation; no evidence was found that the raw initializer was subsequently rescued at fixed aspect ratio.

References

[1] E. J. Candès, X. Li, and M. Soltanolkotabi, *Phase Retrieval via Wirtinger Flow: Theory and Algorithms*, IEEE Trans. Inform. Theory 61 (2015), 1985–2007; arXiv:1407.1065.

[2] Y. Chi, Y. M. Lu, and Y. Chen, *Nonconvex Optimization Meets Low-Rank Matrix Factorization: An Overview*, IEEE Trans. Signal Process. 67 (2019), 5239–5269; arXiv:1809.09573.